

30+ LPA Data Engineer + AI Roadmap (Detailed Learning Plan)

This roadmap includes structured learning phases, important topics, project ideas, and curated learning resources from YouTube, official documentation, O'Reilly, Udemy, Microsoft Learn, and community platforms.

Recommended Learning Order

SQL → Python → PySpark → Spark Internals → Databricks + Delta → Azure + ADF → Streaming + Kafka → System Design → AI Integration → DevOps + Terraform

Suggested Timeline

Phase	Estimated Duration
SQL + Python Refresh	1 Month
PySpark + Spark Internals	2 Months
Databricks + Delta + Azure	2 Months
Streaming + Kafka + CI/CD	1 Month
System Design + AI Integration	1–2 Months
Projects + Interview Preparation	Continuous

1. SQL (Top Priority)

- Joins, CTEs, Subqueries, Aggregations
- Window Functions
- Recursive CTEs
- Query Optimization
- Execution Plans
- Complex ETL Queries
- SQL → PySpark conversions

Learning Resources:

- LeetCode SQL 50
- DataLemur
- SQLBolt
- freeCodeCamp SQL Full Course
- TechTFQ SQL Channel

2. Python for Data Engineering

- Functions and OOP
- Exception Handling
- Iterators and Generators
- Decorators
- Lambda, map/filter/reduce
- JSON/XML parsing
- REST APIs

Learning Resources:

- Corey Schafer Python
 - freeCodeCamp Python
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3. PySpark

- SparkSession
- DataFrames and RDDs
- Transformations vs Actions
- Lazy Evaluation
- DAG
- Window Functions
- Joins
- CSV, JSON, XML, Parquet, Delta

Learning Resources:

- Apache Spark Docs
 - Data With Danny
 - Learn By Example
 - Databricks YouTube
 - Spark Summit Talks
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4. Spark Internals + Optimization

- Driver and Executors
- DAG, Stages, Tasks
- Catalyst Optimizer
- Tungsten
- AQE
- Shuffle Handling
- Partitioning
- Broadcast Joins
- Checkpointing

Learning Resources:

- Jacek Laskowski Spark
 - Databricks Engineering Talks
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5. Databricks + Delta Lake

- Clusters and Jobs
- DBFS
- Mount Points
- ADLS Integration
- MERGE and UPSERT
- VACUUM
- OPTIMIZE
- ZORDER
- Time Travel

Learning Resources:

- Databricks Academy
 - Delta Lake Docs
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6. Azure + ADF

- Storage Accounts
- Blob Storage
- ADLS Gen2
- Azure Key Vault
- ADF Pipelines
- Triggers
- Monitoring
- Azure Synapse
- Azure Monitor

Learning Resources:

- Microsoft Learn Azure DE
 - Adam Marczak Azure
 - AzureLib
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7. Kafka + Streaming

- Structured Streaming
- Watermarking
- Checkpointing

- Kafka Topics
- Partitions
- Offsets
- Consumer Groups

Learning Resources:

- Conduktor Kafka
 - Kafka Docs
 - Stephane Maarek Kafka
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8. DevOps + CI/CD

- Git
- Branching
- PR Reviews
- Azure DevOps
- Build Pipelines
- Release Pipelines
- CI/CD

Learning Resources:

- TechWorld with Nana
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9. Terraform + Docker + Kubernetes

- Terraform Basics
- Azure Infrastructure
- Docker Containers
- Dockerfile
- Kubernetes Pods and Deployments

Learning Resources:

- Terraform Learn
 - Docker Docs
 - Kubernetes Docs
 - TechWorld with Nana Docker/K8s
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10. System Design

- Batch Pipelines
- Streaming Systems
- CDC Pipelines

- Lakehouse Architecture
- Scalability
- Fault Tolerance

Learning Resources:

- Gaurav Sen
 - Design Gurus
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11. AI + LLM Integration

- LLM Basics
- Embeddings
- Vector Databases
- RAG Architecture
- LangChain
- LlamaIndex
- OpenAI APIs
- AI-powered ETL

Learning Resources:

- Andrew Ng AI Courses
 - LangChain Docs
 - LlamaIndex Docs
 - OpenAI Docs
 - freeCodeCamp RAG Course
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12. Projects

- ADF → ADLS → Databricks → Delta → Gold
- Kafka → Spark Streaming → Delta
- RAG Chatbot
- Lakehouse with CI/CD
- Terraform Deployment

Learning Resources:

- GitHub
 - Azure Architecture Center
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Important Career Advice:

Do not only watch tutorials. Build projects continuously, optimize Spark jobs, understand internals deeply, and explain architecture tradeoffs clearly. That is what separates 30+ LPA engineers from average Data Engineering profiles.